

NOATAK, AK, USA

WMO: 701335

Lat: 67.566N

Lon: 162.975W

Elev: 88

StdP: 14.65

Time Zone: -9.00 (NAK)

Period: 95-19

WBAN: 26648

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.3	-35.3	-29.6	1.1	-22.8	-28.7	1.1	-20.7	28.8	19.7	26.2	19.4	1.7	10	0.823

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	75.0	58.5	71.8	56.9	66.8	55.0	60.1	72.1	58.0	68.5	56.1	65.7	7.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.7	63.8	64.0	52.5	58.8	61.5	50.5	54.5	59.1	26.4	72.0	25.0	67.4	23.7	65.5	66.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.1	20.9	18.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-29.4	62.6	3.6	1.7	-32.0	63.8	-34.2	64.8	-36.2	65.7	-38.8	67.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.5	-2.9	3.2	5.2	17.7	36.9	52.4	54.0	49.6	41.6	24.9	7.2	2.6
	DBStd	24.06	19.31	17.15	12.77	11.98	10.48	7.76	6.27	5.64	7.17	11.48	13.28	15.55
	HDD50	9688	1639	1311	1389	968	418	63	31	74	261	779	1285	1470
	HDD65	14796	2104	1731	1854	1418	871	383	342	478	702	1243	1735	1935
	CDD50	376	0	0	0	0	12	135	157	62	9	1	0	0
	CDD65	9	0	0	0	0	0	5	3	1	0	0	0	0
	CDH74	157	0	0	0	0	3	63	71	13	0	7	0	0
	CDH80	21	0	0	0	0	0	8	9	0	0	4	0	0
Wind	WSAvg	7.5	6.9	8.6	7.8	8.3	7.4	6.9	6.5	7.0	7.5	7.8	7.2	7.7
Precipitation	PrecAvg	17.8	1.3	1.4	1.0	0.8	0.7	0.8	1.9	2.8	2.0	2.4	1.5	1.2
	PrecMax	24.8	2.7	4.4	5.0	2.2	1.8	1.5	3.9	6.1	4.4	4.6	4.6	3.1
	PrecMin	12.8	0.3	0.2	0.2	0.1	0.0	0.4	0.5	0.6	0.5	0.9	0.2	0.4
	PrecStd	3.1	0.6	1.0	0.9	0.6	0.5	0.3	0.8	1.3	0.8	0.9	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	36.3	35.5	46.0	72.0	80.7	81.1	75.0	63.2	53.9	36.4	36.7
		MCWB	32.7	32.8	32.7	38.3	53.4	59.5	61.7	61.5	51.5	42.4	33.2	32.6
	2%	DB	31.5	32.5	31.6	40.9	63.7	75.0	75.2	66.5	58.7	44.8	33.9	33.8
		MCWB	29.7	30.8	29.8	35.0	50.2	57.3	59.4	56.3	48.6	39.7	31.6	31.0
	5%	DB	28.1	29.8	28.0	37.1	58.6	72.0	71.9	63.3	55.0	39.9	30.1	28.4
		MCWB	26.3	28.5	25.4	32.8	47.2	55.9	58.2	54.0	47.8	36.0	29.0	26.7
	10%	DB	24.9	27.2	25.2	35.0	53.7	67.8	67.5	60.6	52.4	37.4	26.9	24.0
		MCWB	22.8	25.3	22.8	31.7	44.9	54.0	57.0	52.6	46.9	33.9	25.0	22.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.5	33.1	33.1	38.6	54.3	61.4	62.7	62.9	53.1	44.3	33.9	33.7
		MCDB	35.0	35.2	34.7	45.7	69.2	77.2	76.4	74.6	58.3	51.0	36.0	36.1
	2%	WB	29.9	31.5	29.5	35.6	50.7	58.6	60.7	57.9	51.0	40.1	31.6	30.9
		MCDB	31.8	33.2	31.5	40.1	62.9	72.3	72.9	64.5	55.7	44.6	33.0	32.8
	5%	WB	27.5	28.5	25.7	33.6	47.8	56.7	59.0	55.6	49.4	37.5	28.7	27.2
		MCDB	28.8	29.4	27.8	37.0	58.1	69.4	70.1	62.0	54.6	39.8	29.9	29.1
	10%	WB	24.1	25.7	23.1	31.7	44.9	54.5	57.0	53.2	47.3	34.8	25.0	22.8
		MCDB	26.3	27.4	25.7	34.7	52.7	66.5	66.2	59.0	51.9	37.5	26.8	24.9
Mean Daily Temperature Range	5% DB	MDBR	14.4	14.8	16.6	16.5	16.4	19.1	17.5	15.0	14.4	11.7	11.7	13.8
		MCDBR	14.1	13.1	15.8	15.6	25.7	24.5	27.2	21.6	18.7	12.9	9.6	13.5
		MCWBR	12.7	12.5	14.0	11.8	14.9	11.9	14.1	13.1	11.8	9.3	8.8	11.9
	5% WB	MCDBR	13.5	13.3	14.5	14.6	24.5	22.5	23.6	18.7	15.8	12.9	9.7	12.6
		MCWBR	12.4	12.8	13.0	11.3	14.5	11.9	12.8	11.8	10.3	9.3	9.1	11.3
Clear-Sky Solar Irradiance	taub	0.174	0.241	0.261	0.325	0.369	0.341	0.348	0.357	0.309	0.245	0.168	N/A	
	taud	1.903	2.088	2.226	2.096	2.165	2.443	2.450	2.424	2.502	2.317	1.942	N/A	
	Ebn at Noon	49	183	252	258	259	274	268	250	238	188	48	0	
	Edh at Noon	3	16	24	37	39	30	29	27	19	12	3	0	
All-Sky Solar Radiation	RadAvg	16	184	742	1443	1748	1900	1512	1112	681	246	39	0	
	RadStd	2	22	62	81	200	174	163	141	80	28	5	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+2.57	N/A	N/A	+1.80	+1.48	N/A	-855	-918	+117	+2

Nomenclature: See separate page